



Lesson 3.3 — Zeros and Multiplicity

Big idea: Where a polynomial crosses (or touches) the x-axis is determined by its zeros and how many times each one is repeated.

Quick review

Zero: any x with $P(x) = 0$. From factored form $a(x - r)^k(\dots)$, the zeros are the r values.

Multiplicity: the exponent k on $(x - r)$. It tells you how the graph behaves at that zero.

Odd multiplicity: graph **CROSSES** through the x-axis at that zero.

Even multiplicity: graph **TOUCHES** the x-axis and bounces back without crossing.

Total zeros: with multiplicity counted, equals the degree of the polynomial (over complex numbers).

Worked example

Worked example. $P(x) = (x + 2)(x - 3)^2(x - 1)^3$. Find zeros, multiplicities, and behavior at each.

Zero $x = -2$ with mult. 1 (crosses). Zero $x = 3$ with mult. 2 (bounces). Zero $x = 1$ with mult. 3 (crosses, with a flatter shape).

Warm-ups

1. Find the zeros of $P(x) = (x - 4)(x + 1)(x - 7)$.
2. What is the multiplicity of $x = 5$ in $P(x) = (x - 5)^2(x + 3)$?
3. How does the graph of $P(x) = x^2(x - 2)$ behave at $x = 0$?

Practice

1. Find all zeros and their multiplicities for $P(x) = x^3(x + 1)^2(x - 4)$.
2. Write a polynomial of degree 4 with zeros 0 (multiplicity 2) and ± 3 (each simple).
3. $P(x) = (x - 1)^2(x + 2)$. Sketch end behavior plus the zero behavior in words.

Challenge

1. A degree-4 polynomial has zeros at $x = 1$, $x = -2$ (multiplicity 2), and $x = 5$. Write a possible equation in factored form.
2. If a polynomial has zeros 2, $1+i$, and $1-i$ (all simple), what is its minimum possible degree, and why must the complex zeros come in a conjugate pair (assuming real coefficients)?



Answer key — Lesson 3.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 4$, $x = -1$, $x = 7$.
2. Multiplicity 2.
3. $x = 0$ has multiplicity 2 (even) $\rightarrow$ graph TOUCHES and bounces.

Practice

1. $x = 0$ (mult 3, crosses), $x = -1$ (mult 2, bounces), $x = 4$ (mult 1, crosses). Total degree 6.
2. $P(x) = a x^2(x - 3)(x + 3)$; pick $a = 1 \rightarrow x^4 - 9x^2$.
3. Degree 3, leading $+$ $\rightarrow$ down/up. At $x = 1$, mult 2 $\rightarrow$ bounces. At $x = -2$, mult 1 $\rightarrow$ crosses.

Challenge

1. $P(x) = a(x - 1)(x + 2)^2(x - 5)$. Common simple choice: $a = 1$.
2. Minimum degree 3. Real-coefficient polynomials must have complex zeros in conjugate pairs because complex zeros of real polynomials always come in conjugate pairs (a consequence of the conjugate root theorem). So $1+i$ and $1-i$ must come together.